# Gully Expansion and Erosion Rates on Lakes Huron and Erie

Jess Zhang
Department of Earth and Environmental Sciences, University of Waterloo
Waterloo, Ontario, Canada
jess.zhang@uwaterloo.ca

LAKE HURON COASTAL CENTRE

UNIVERSITY OF WATERLOO | FACULTY OF SCIENCE Department of Earth and Environmental Sciences

## Motivation

- Erosion on the coasts of Lakes Erie and Huron poses risks to critical infrastructure and coastal communities. Sandy bluffs on the north shore of Lake Erie retreat at 1 to 15 m $y^{-1}$ [1], while clayey bluffs on the east shore of Lake Huron retreat at less than 1 m $y^{-1}$ [2].
- Gullying can erode bluff material much faster, but gully erosion rates have not been quantified. Gully area growth was last quantified in 1995 [3].
- Gullies form at topographic highs when water channelizes and cuts down through weak geology. Agricultural deforestation is the suspected cause of gully initiation [3].
- Quantifying gully erosion volume rates and areal growth rates is necessary to predict erosion and inform coastal land use planning.

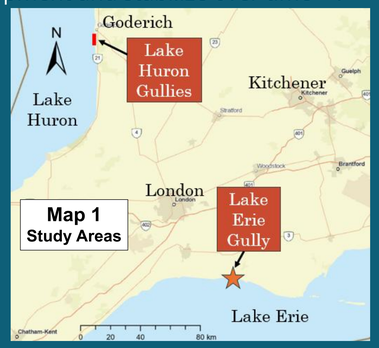



## Objectives

- Quantify rates of gully areal growth.
- Quantify rates of gully erosion, i.e., quantity of sediment being added to the littoral zone.
- Understand how and where gullies develop.

## Background

- Coastal erosion along Lake Erie and Lake Huron is driven by elevated lake levels and reduced winter ice cover, accelerating bluff retreat and often starving downdrift features [1,2]. Increased erosion correlates with coarser materials and shoreline modifications, as seen near Port Burwell [1].
- On Lake Huron, coastal gullies show consistent areal growth, where higher growth rates are sustained by larger watersheds. Initiation is linked to agricultural deforestation while expansion is influenced by snowmelt and drainage redirection [3].
- Gully growth rates are highest in early development and stabilize over time [4].




## Preliminary Results

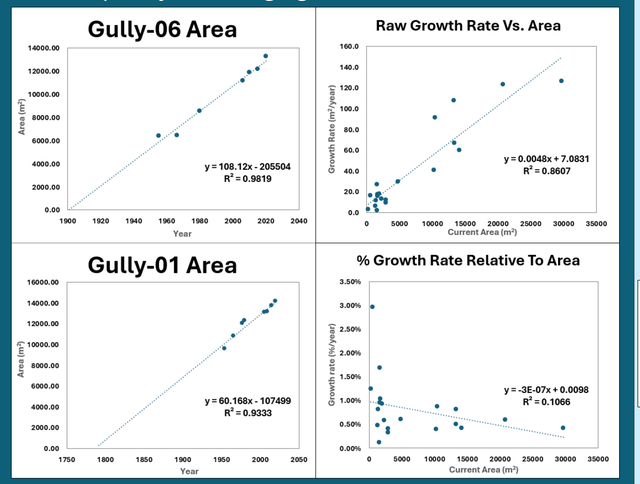
- All gullies are growing consistently barring rehabilitation, confirming previous studies [3].

### Erosion Rates

- On Lake Huron, the fastest-growing gully eroded 1287 $m^3$ per year between 1955 and 2020.
- The Port Burwell gully eroded 27669 $m^3$ per year between 2013 and 2015, or 83008 $m^3$ total.
- This is 2% of the total sediment volume added to the littoral cell from Port Burwell to Long Point between 2010 and 2015 (3,655,000 $m^3$) [1].

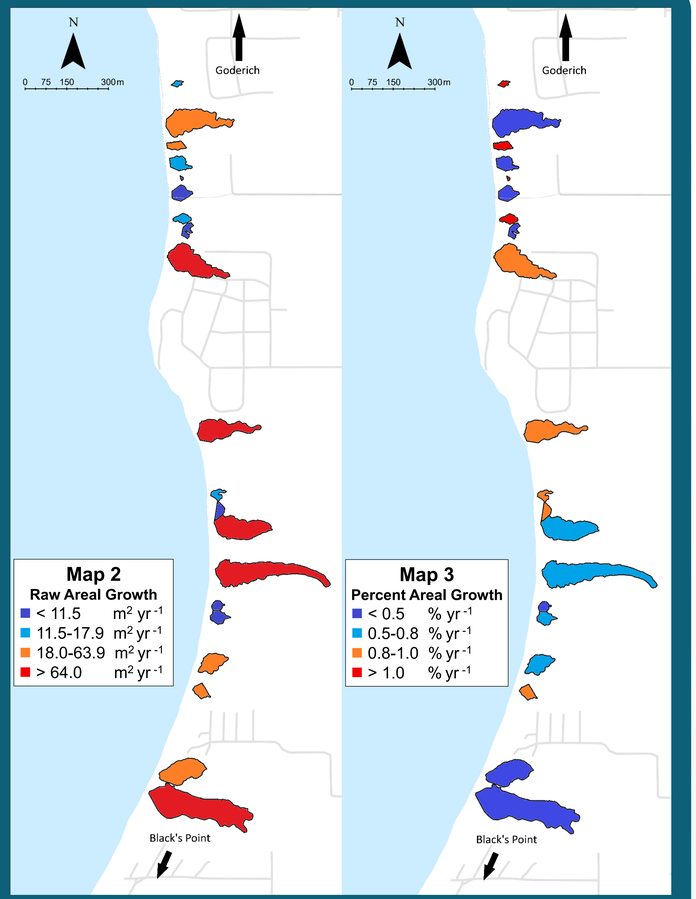
### Areal Expansion

- On Lake Huron, the fastest-growing gully grew at 127 $m^2$ per year between 1955 and 2020.
- The Port Burwell gully grew at 4403 $m^2$ per year between 2013 and 2020.
- Large gullies grow more quickly in raw area and volume.
- Proportionally to their size, small gullies grow in area more quickly than large gullies.







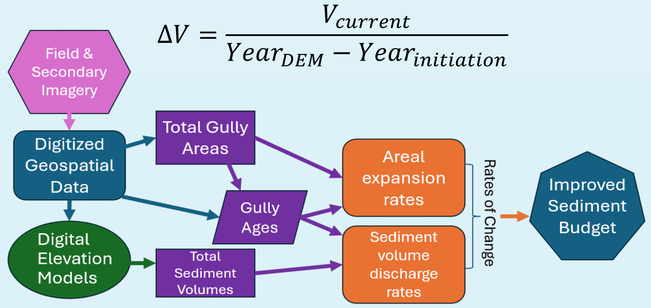



| Lake | Gully | Areal Growth Rate | | Erosion Rate | |
|---|---|---|---|---|---|
| | | $m^2$ $yr^{-1}$ | # parking spots $yr^{-1}$ | $m^3$ $yr^{-1}$ | # garbage trucks $yr^{-1}$ |
| Erie | Port Burwell | 4403 | 367 | 27669 | 1537 |
| Huron | Gully-15 | 127 | 11 | 1287 | 71 |
| | Gully-11 | 13 | 1 | 115 | 6 |
| | Average | 40 | 3 | 413 | 23 |

## Methods (Continued)

1. Field Data Collection: Programmed **UAV (unmanned aerial vehicle) surveys** were conducted to capture plan-view images with 70 to 80% overlap, at a height 60 m above the bluffs.
2. Secondary Data Collection: **High-resolution imagery and elevation data** between 1931 and 2020 was downloaded from public sources or purchased from libraries.
3. Geospatial Data Processing:
   **Gully areas** were quantified via **digitization** in ArcGIS for 1955, 1966, 1978, 1980, 2006, 2010, 2015, and 2020.
   **Digital Elevation Models (DEMs)** were constructed from 2025 field data, using the Structure from Motion technique in Pix4D.
   **Gully volumes** were calculated from **DEMs** from 2015, 2020, and 2025 using the Surface Volume tool in ArcGIS.
4. Linear Regression:
   **Annual areal growth rates** were determined using the SLOPE() function in Excel, for data from 1955 to 2020.
   **Gully initiation years** were back-dated using the INTERCEPT() function in Excel, projecting the best-fit line backwards to find the year where it intersects the horizontal years axis.
5. **Annual erosion rates** were determined by averaging the most recent available gully volume across the length of the gully's life since initiation.

$$\Delta V = \frac{V_{current}}{Year_{DEM} - Year_{initiation}}$$




## Conclusion & Recommendations

Gullies grow fastest during their initial years, and larger gullies grow faster compared to small gullies. Large gullies should be prioritized when monitoring for threats to infrastructure and sediment contributions to the littoral cell. New, large gullies (i.e., Port Burwell) should be monitored closely in initial years, but will not expand indefinitely at their initial rate. Continued digitization and UAV-based sediment volume analysis are essential to fully characterize erosional contributions of bluff gullies and inform coastal land use management along both lakeshores.

## Acknowledgements & References

Many thanks to Dr. Chris Houser for his thoughtful guidance, to the Lake Huron Coastal Centre for funding, and to Ben Woodward and Cooper O'Rourke for continual support throughout the project.

[1] O'Rourke, C. et al. (2025). *Rates of Bluff Retreat and Shoreline Change Along the North Shore of Lake Erie*. Journal of Great Lakes Research, 102703. https://doi.org/10.1016/j.jglr.2025.102703 [2] Woodward, B. et al. (in prep). *Using Community Archives to Understand Long-Term Coastal Change: A Case Study from Southeastern Lake Huron*. [3] Burkard, M. B., & Kostaschuk, R. A. (1995). Initiation and evolution of gullies along the shoreline of Lake Huron. Geomorphology, 14(3), 211-219. https://doi.org/10.1016/0169555X(95)00059-E [4] Sidorchuk, A. (1999). Dynamic and static models of gully erosion. CATENA, 37(3-4), 401-414. https://doi.org/10.1016/S0341-8162(99)00029-6